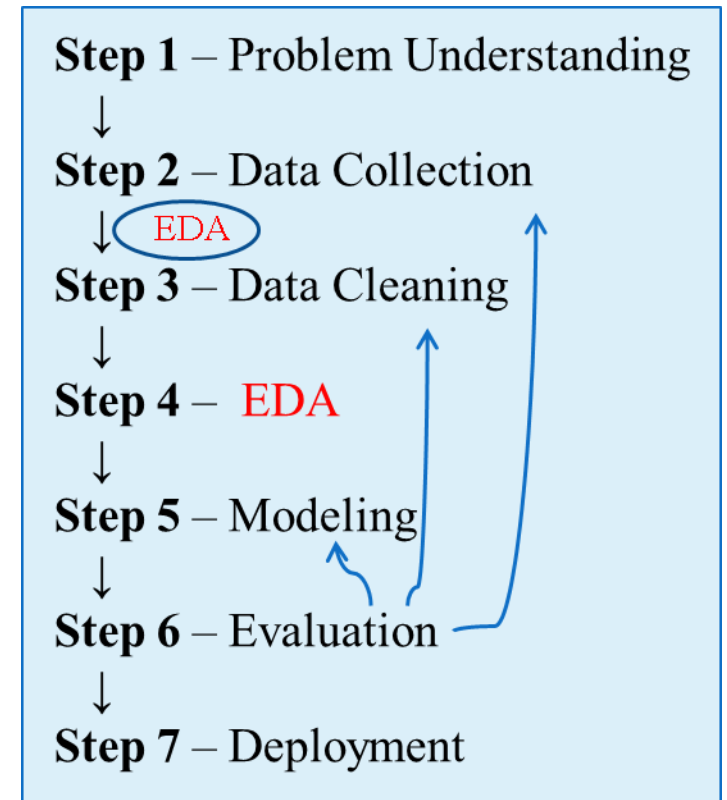


# Exploratory Data Analysis (EDA)

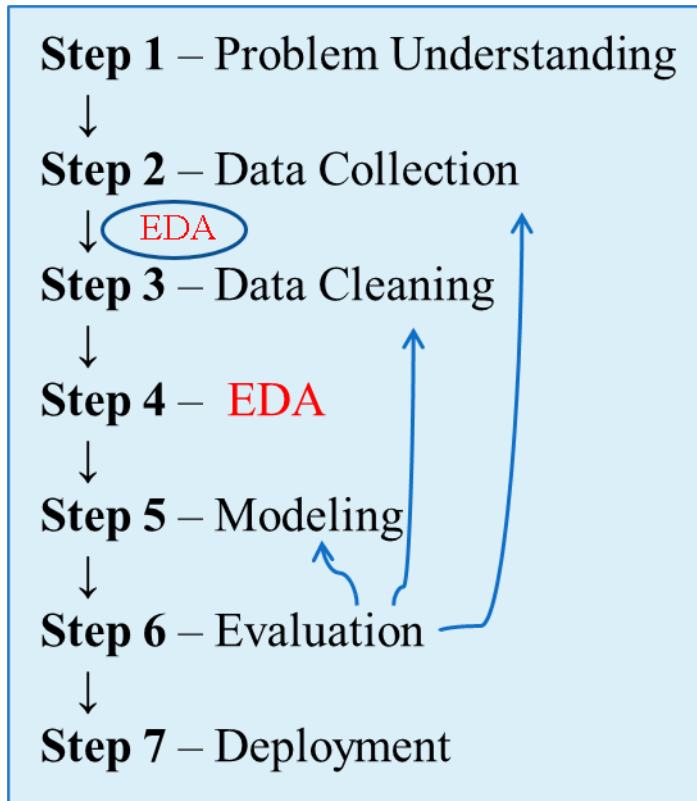
Exploratory Data Analysis (EDA) is the process of understanding a dataset before applying statistical models or machine learning algorithms.

EDA answers the question:

What does my data look like, and is it suitable for analysis?



# Exploratory Data Analysis (EDA)



A Typical EDA Workflow

1. Load Dataset  
↓
2. Understand Structure  
↓
3. Check Variable Types  
↓
4. Compute Summary Statistics  
↓
5. Identify Missing Values  
↓
6. Detect Duplicates  
↓
7. Detect Outliers  
↓
8. Visualize Distributions  
↓
9. Examine Relationships  
↓
10. Generate Insights  
↓
11. Prepare for Data Cleaning/Modeling

# Understanding Data

Before analyzing data, how do we understand what data we have?

| StudentID | Age | CGPA | Department |
|-----------|-----|------|------------|
| 101       | 21  | 3.75 | CSE        |
| 102       | 22  | 3.20 | EEE        |

Given the example dataset:

- What information can we obtain from this data?
- Is this enough to understand all students?
- What should we examine before applying machine learning?

# Understanding Data

## Question 1: What information can we obtain from this data?

We can identify:

- Student IDs
- Students' ages
- Students' CGPAs
- Departments they belong to

We can also perform simple analyses such as:

- Average age
- Average CGPA
- Highest and lowest CGPA
- Number of students in each department

Data contains information that can be summarized and analyzed to answer questions.

# Understanding Data

**Question 2: Is this enough to understand all students?**

**Answer: No**

Reasons:

- Only two students are shown.
- The dataset may not represent the entire student population.
- Important variables may be missing:
  - Gender
  - Semester
  - Attendance
  - Exam scores
  - Family background
  - Extracurricular activities

**Population vs Sample: A small dataset may provide insights, but it may not fully represent all students.**

# Understanding Data

## Question 3: What should we examine before applying machine learning?

Before building a machine learning model, we should examine:

### 1. Data Quality

- Missing values
- Incorrect values
- Duplicate records

Example: Age = -5, CGPA = 12.5 **Clearly invalid.**

### 2. Data Types

Determine whether variables are:

- Numerical
- Categorical

Example:

| Variable   | Type        |
|------------|-------------|
| Age        | Numerical   |
| CGPA       | Numerical   |
| Department | Categorical |

# Understanding Data

## **Question 3: What should we examine before applying machine learning?**

Before building a machine learning model, we should examine:

### **3. Dataset Size**

- How many rows?
- How many features?

A model may not perform well with very little data.

### **4. Distribution of Variables**

Questions such as:

- Are ages normally distributed?
- Are CGPAs concentrated in a narrow range?

# Understanding Data

## Question 3: What should we examine before applying machine learning?

Before building a machine learning model, we should examine:

### 5. Missing Values

- Example:

| Age | CGPA |
|-----|------|
| 21  | 3.5  |
| NA  | 3.7  |

Need handling before modeling.

### 6. Outliers

Example: CGPA

3.2

3.5

3.7

15.0

Clearly problematic.

# Understanding Data

## **Question 3: What should we examine before applying machine learning?**

Before building a machine learning model, we should examine:

### **7. Target Variable Availability**

For supervised learning:

- What are we trying to predict?

Examples:

- Pass/Fail
- CGPA
- Placement Status

**Without a target variable, many ML tasks cannot be performed.**

# Understanding Data: Structured vs. Unstructured Data

## Structured Data

- Organized in rows and columns
- Stored in databases, spreadsheets

| ID | Name  | CGPA |
|----|-------|------|
| 1  | Rahim | 3.7  |

## Unstructured Data

- No predefined format

### Examples:

- Emails
- PDFs
- Images
- Audio
- Videos
- Social media posts

## Semi-Structured Data

### Examples:

- JSON
- XML

### Which type is:

- Facebook post?
- Hospital record?
- YouTube video?
- Student database?

# Understanding Data: Types of Variables

Variables are the characteristics measured for each observation.

## Categorical Variables

### Nominal

Examples:

- Gender
- Blood Group
- Department

### Ordinal

Examples:

- Satisfaction Level
- Education Level
- Product Ratings

## Numerical Variables

### Discrete

Examples:

- Number of students
- Number of courses

### Continuous

Examples:

- Height
- Weight
- Income

| Variable     | Type |
|--------------|------|
| Age          | ?    |
| CGPA         | ?    |
| Department   | ?    |
| Rating (1–5) | ?    |

# Understanding Datasets

## Observation

A row

## Feature / Attribute

An input column

Examples:

- Age
- CGPA

| ID | Age | CGPA | Pass |
|----|-----|------|------|
| 1  | 20  | 3.5  | Yes  |
| 2  | 21  | 3.2  | No   |

## Label / Target

Output variable

Example: Pass

## Dataset Dimensions

Rows  $\times$  Columns

Example:  $1000 \times 20$

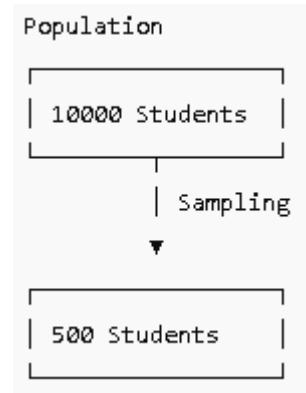
# Population vs Sample

Population: All AIUB students

Sample: 500 selected students

## Why Sampling?

- Cost
- Time
- Practicality



Can conclusions from a sample be trusted? Lead into statistical thinking.

# Descriptive Statistics

**Descriptive statistics** is the process of condensing key characteristics of the dataset into simple numeric metrics. Some of the common quantitative metrics used are mean, standard deviation, and correlation.

Descriptive statistics can be broadly classified into

- Univariate Exploration: Involves one variable
- Multivariate Exploration: Involves more than one variable

# Descriptive Statistics: Measures of Central Tendency

## What is Central Tendency?

Measures of Central Tendency describe the **center** or **typical value** of a dataset.

The three most common measures are:

- **Mean** (Average)
- **Median** (Middle Value)
- **Mode** (Most Frequent Value)

## Example Dataset

Suppose the quiz marks of 7 students are:

| Student | Marks |
|---------|-------|
| S1      | 70    |
| S2      | 80    |
| S3      | 75    |
| S4      | 90    |
| S5      | 75    |
| S6      | 85    |
| S7      | 75    |

# Descriptive Statistics: Measures of Central Tendency

Dataset: 70, 80, 75, 90, 75, 85, 75

## Mean (Average)

Sum of all marks:  $70 + 80 + 75 + 90 + 75 + 85 + 75 = 550$

Number of students:  $n = 7$

Mean:  $550 / 7 = 78.57$

## Interpretation

On average, students scored approximately **78.57 marks**.

| Student | Marks |
|---------|-------|
| S1      | 70    |
| S2      | 80    |
| S3      | 75    |
| S4      | 90    |
| S5      | 75    |
| S6      | 85    |
| S7      | 75    |

# Descriptive Statistics: Measures of Central Tendency

## Median (Middle Value)

First, sort the data: 70, 75, 75, 75, 80, 85, 90

The middle observation is: 75

**Median = 75**

## Interpretation

Half of the students scored below 75 and half scored above 75.

| Student | Marks |
|---------|-------|
| S1      | 70    |
| S2      | 80    |
| S3      | 75    |
| S4      | 90    |
| S5      | 75    |
| S6      | 85    |
| S7      | 75    |

# Descriptive Statistics: Measures of Central Tendency

## Mode (Most Frequent Value)

Count frequencies:

| Marks | Frequency |
|-------|-----------|
| 70    | 1         |
| 75    | 3         |
| 80    | 1         |
| 85    | 1         |
| 90    | 1         |

The most frequent value is:75

Mode = 75

## Interpretation

The most common score obtained by students is 75.

| Student | Marks |
|---------|-------|
| S1      | 70    |
| S2      | 80    |
| S3      | 75    |
| S4      | 90    |
| S5      | 75    |
| S6      | 85    |
| S7      | 75    |

# Descriptive Statistics: Measure of Spread

- Summarizing the dataset can help us understand the data, especially when the dataset is large.
- The mode, median, and mean **summarize the data into a single value** that is typical or representative of all the values in the dataset, but this is only part of the 'picture' that summarizes a dataset.
- **Measures of spread describe how similar or varied the set of observed values** are for a particular variable (data item).
- Measures of spread summarize the data in a way that shows how scattered the values are and how much they differ from the mean value.

# Descriptive Statistics: Measure of Spread

| Dataset A                          | Dataset B                            |
|------------------------------------|--------------------------------------|
| 4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8 | 1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11 |

The mode (most frequent value), median (middle value\*) and mean (arithmetic average) of both datasets is 6. *If we just looked at the measures of central tendency, we may assume that the datasets are the same.*

However, if we look at the spread of the values in the following graph, we can see that Dataset B is more dispersed than Dataset A. Used together, *the measures of central tendency and measures of spread help us to better understand the data.*

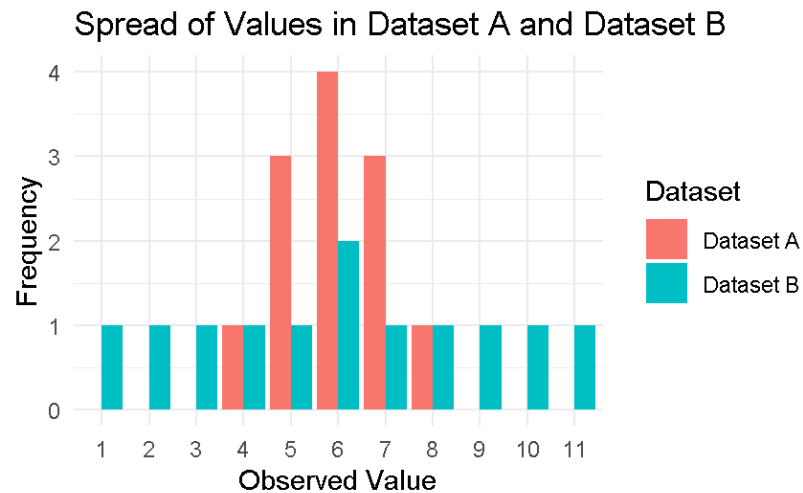
*Two datasets can have the same mean, median, and mode, yet be very different.*

| Measure | Dataset A | Dataset B |
|---------|-----------|-----------|
| Mean    | 6         | 6         |
| Median  | 6         | 6         |
| Mode    | 6         | 6         |
| Range   | 4         | 10        |
| SD      | 1.13      | 3.16      |

# Descriptive Statistics: Measure of Spread

| Dataset A                          | Dataset B                            |
|------------------------------------|--------------------------------------|
| 4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8 | 1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11 |

| Measure | Dataset A | Dataset B |
|---------|-----------|-----------|
| Mean    | 6         | 6         |
| Median  | 6         | 6         |
| Mode    | 6         | 6         |
| Range   | 4         | 10        |
| SD      | 1.13      | 3.16      |



Measure of spread, sometimes also called a measure of dispersion, is used to describe the variability in a sample or population. **There are two common metrics to quantify spread: Range and Deviation.**

**Range:** The range is the difference between the maximum value and the minimum value of the attribute.

# Descriptive Statistics: Measure of Spread

**Deviation:** The **variance** and **standard deviation** measures the spread, by considering all the values of the attribute. Deviation is simply measured as the difference between any given value ( $x_i$ ) and the mean of the sample ( $\mu$ ). The **variance** is the sum of the squared deviations of all data points divided by the number of data points. For a dataset with  $N$  observations, the variance is given by the following equation.

$$\text{Variance} = s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Standard deviation is the square root of the variance. Standard deviation measures how much your entire data set differs from the mean.

High standard deviation means the data points are spread widely around the central point. Low standard deviation means data points are closer to the central point.

## An example with details

Imagine we have a dataset of exam scores of 10 students:

### Descriptive Statistics

**Mean** (Average):

$$(85+78+92+67+88+75+90+82+79+95)/10$$
$$=831/10=83.1$$

**Median:**

Ordered scores: 67, 75, 78, 79, 82, 85, 88, 90, 92, 95

Median = Average of 5th and 6th values =  $(82 + 85) / 2 = 83.5$

Minimum Score: 67, Maximum Score: 95

**Range:**  $95 - 67 = 28$

| Student | Score |
|---------|-------|
| A       | 85    |
| B       | 78    |
| C       | 92    |
| D       | 67    |
| E       | 88    |
| F       | 75    |
| G       | 90    |
| H       | 82    |
| I       | 79    |
| J       | 95    |

## An example with details

### Variance:

First calculate mean = 83.1

Then compute squared deviations from the mean:

$$(85 - 83.1)^2 = 3.61, (78 - 83.1)^2 = 26.01, (92 - 83.1)^2 = 79.21$$

$$(67 - 83.1)^2 = 259.21, (88 - 83.1)^2 = 24.01, (75 - 83.1)^2 = 65.61$$

$$(90 - 83.1)^2 = 47.61, (82 - 83.1)^2 = 1.21, (79 - 83.1)^2 = 16.81,$$

$$(95 - 83.1)^2 = 141.61$$

$$\text{Sum} = 664.9$$

$$\text{Variance} = 664.9 / 9 \approx 73.88$$

$$\text{Standard Deviation: } \sqrt{73.88} \approx 8.6$$

| Student | Score |
|---------|-------|
| A       | 85    |
| B       | 78    |
| C       | 92    |
| D       | 67    |
| E       | 88    |
| F       | 75    |
| G       | 90    |
| H       | 82    |
| I       | 79    |
| J       | 95    |

## Exercise 1

You have two datasets representing the monthly test scores (out of 100) for students in two different classes:

**Class X:** 75, 78, 80, 82, 85, 87

**Class Y:** 60, 65, 75, 80, 90, 95

### Tasks:

1. Calculate the range, variance, and standard deviation for each class.
2. Which class shows more variability in test scores?
3. What might this indicate about the performance consistency among students in each class?

## Exercise 2

You are given the following dataset representing the number of hours 10 employees worked last week:

40, 42, 38, 41, 44, 39, 40, 37, 60, 38

### Tasks

1. Calculate the mean and median of the dataset.
2. Which measure (mean or median) better represents the central tendency of the data?
3. Justify your answer considering the effect of any outlier(s).

# Descriptive Statistics: Multivariate

Multivariate Data Exploration refers to the process of analyzing **more than two variables simultaneously** to understand relationships, patterns, or structures in a dataset.

It's an important step in exploratory data analysis (EDA), especially when working with real-world datasets where multiple features interact.

## Why Multivariate Analysis?

- To discover interdependencies between variables.
- To identify patterns or trends that are not obvious from univariate or bivariate analysis.
- To reduce dimensionality or detect redundant or highly correlated features.
- To prepare the dataset for predictive modeling.

# Descriptive Statistics: Multivariate

## Why Multivariate Statistics?

Many real-world questions involve relationships between variables.

Examples:

| Question                                   | Variables                       |
|--------------------------------------------|---------------------------------|
| Does study time affect CGPA?               | Study Hours, CGPA               |
| Does age influence salary?                 | Age, Salary                     |
| Does advertising affect sales?             | Advertising Cost, Sales         |
| Does temperature affect electricity usage? | Temperature, Energy Consumption |

# Descriptive Statistics: Multivariate

**Covariance:** Covariance measures whether two variables tend to move together.

## Interpretation

| Covariance | Meaning                                 |
|------------|-----------------------------------------|
| Positive   | Variables increase together             |
| Negative   | One increases while the other decreases |
| Near Zero  | Little linear relationship              |

**Example Dataset →**

**Observation:**

As Study Hours increase, CGPA also increases.

Therefore: Covariance  $> 0$

| Student | Study Hours (X) | CGPA (Y) |
|---------|-----------------|----------|
| 1       | 10              | 2.8      |
| 2       | 15              | 3.0      |
| 3       | 20              | 3.2      |
| 4       | 25              | 3.6      |
| 5       | 30              | 3.9      |

# Descriptive Statistics: Multivariate

**Correlation:** Covariance is difficult to interpret because its magnitude depends on units. Correlation solves this problem.

Correlation measures the strength and direction of a relationship.

Range:  $-1 \leq r \leq +1$

| Correlation | Interpretation                 |
|-------------|--------------------------------|
| +1          | Perfect positive relationship  |
| +0.8        | Strong positive relationship   |
| +0.5        | Moderate positive relationship |
| 0           | No linear relationship         |
| -0.5        | Moderate negative relationship |
| -0.8        | Strong negative relationship   |
| -1          | Perfect negative relationship  |

| Study Hours | CGPA |
|-------------|------|
| 10          | 2.8  |
| 15          | 3.0  |
| 20          | 3.2  |
| 25          | 3.6  |
| 30          | 3.9  |

**Strong positive correlation.**

# Descriptive Statistics: Multivariate

**Remember:** *Correlation does not mean causation.* The phrase “correlation does not imply causation” is often used in statistics to point out that correlation between two variables does not necessarily mean that one variable causes the other to occur.

*Example 1: Ice Cream Sales & Shark Attacks ?.*

*Example 2: High School Graduates vs. Pizza Consumption ?*

*Example 3: Master's Degrees vs. Box Office Revenue ?*

# Descriptive Statistics: Multivariate

## Cross-Tabulation (Categorical Variables)

Used when both variables are categorical.

- Which group has higher pass rates?
- Is there a relationship between Gender and Pass Status?

| Gender | Yes | No |
|--------|-----|----|
| Male   | 1   | 1  |
| Female | 2   | 0  |

| Gender | Passed |
|--------|--------|
| Male   | Yes    |
| Female | Yes    |
| Male   | No     |
| Female | Yes    |

# Descriptive Statistics: Multivariate

## Grouped Descriptive Statistics

- Instead of computing one mean for everyone, compute means for groups.
- This provides richer information than a single overall mean.

| Department | CGPA |
|------------|------|
| CSE        | 3.7  |
| CSE        | 3.5  |
| EEE        | 3.1  |
| EEE        | 3.3  |

| Department | Average CGPA |
|------------|--------------|
| CSE        | 3.6          |
| EEE        | 3.2          |